STE 80898–0

TCRseries
Industrial Robot

TCR5L/TS2000

INSTRUCTION MANUAL

MAINTENANCE

Notice

1. Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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TOSHIBA MACHINE CO., LTD.

NUMAZU, JAPAN

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PREFACE

This manual describes how to maintain the robot and the controller.
Maintenance and inspection are essential to keep the robot performance for a long time, to prevent failures and to assure safety. Be sure to read through this manual and set up a maintenance program before actually starting the robot.

This manual is composed of the following sections.

Section 1: Maintenance Items
This section describes the maintenance schedule of the robot and controller and items required for inspection and maintenance.

Section 2: Maintenance of the Robot
This section refers to the robot structure and items required for inspection and maintenance of the robot.

Section 3: Maintenance of Controller
This section refers to the robot controller structure and items required for inspection and maintenance of the controller.

Section 4: Home Point Setting
This section describes each axis home point setting procedures of the robot.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
! **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
! **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger or caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the Safety Manual provided separately before starting actual work to assure safety of the work covering from the robot installation to operation.

[Maintenance and inspection]

Be sure to observe the following items to use the robot safely.

DANGER	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to turn off the main power switch of the controller before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

CAUTION	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use Toshiba Machine's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

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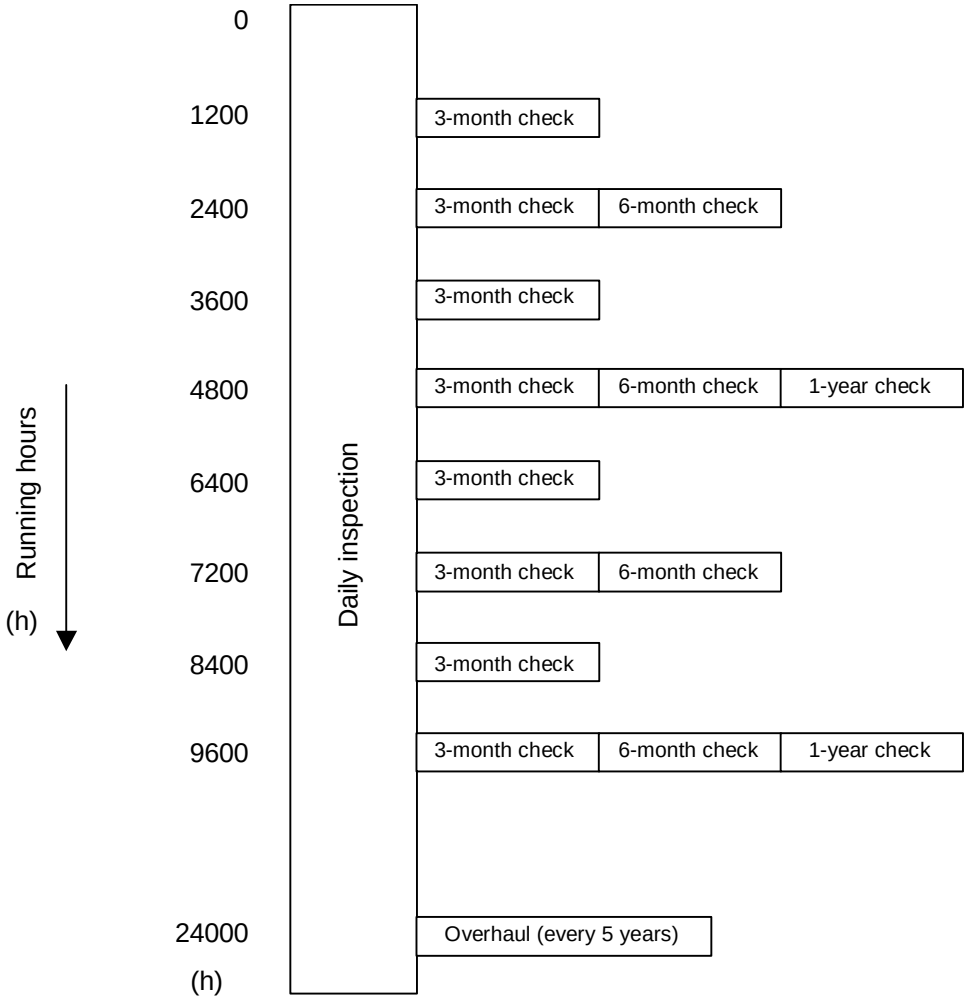
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Section 1 Maintenance Items

1.1 Maintenance Schedule

Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule (yardstick)

- When the system is operated for 16 hours per day:
  16 hours × 25 days × 3 months = 1,200 hours
- When the system is operated for 24 hours per day:
  24 hours × 25 days × 3 months = 1,800 hours

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection. For the execution procedures, see the relevant paragraph listed in the table below.

a) Daily inspection

No.	Item	Inspection procedures	Relevant Para.
1	Abnormal state of cable connector (Perform check before operation.)	Make sure that all cable connectors are tightened completely. Before the robot operation, make sure that each cable is not twisted, bent or cracked locally.	2.3.1
2	Set and clamp screws and bolts of each part (Perform check before operation.)	• Make sure that the robot base installation bolts are tightened completely. • Make sure that the cover clamp screws are tightened completely. • Make sure that the hand clamp bolts are tightened completely.	2.3.2
3	Grease leakage from robot body (Perform check before operation.)	Make sure that no grease leaks from the robot body (i.e., each joint and grease nipple).	
4	Vibration and noise	Make sure that each axis can move smoothly.	
5	Change in positioning accuracy	Make sure that the robot can stop at a predetermined position accurately.	
6	Air leakage (Perform check with noise.)	Make sure that air will not leak from inside the robot body or air joint.	
7	Controller vent holes	(1) Make sure that no obstacle stands in front of the vent holes. (2) Make sure that no contaminant is left in the vent hole.	3.3.1

No.	Item	Inspection procedures	Relevant Para.
8	Function of safety devices	Make sure that each safety device connected to the emergency stop pushbutton switch and external operation input signal can function properly.	3.3.2

b) Quarterly (every three (3)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Abnormal state of cable duct	Make sure that the cable duct is tightened completely. Before the robot operation, make sure that the cable duct is not twisted, bent or cracked locally.	2.3.1
2	Set and clamp screws and bolts of each part (Perform check on complete tightening.)	• Make sure that the robot base installation bolts are tightened completely. • Make sure that the cover clamp screws are tightened completely. • Make sure that the hand clamp bolts are tightened completely.	2.3.3
3	Make sure that the cables are not worn out.	Make sure that the cables of each unit are not worn out.	2.3.4
4	Lubricating condition of the racks and the pinion	Make sure the racks and the pinion are greased sufficiently and that no rust development is found on them.	2.3.5

c) Semi-annual (every six (6)-month) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Each axis drive system	Move each axis by hand and make sure that it moves smoothly. (For the axis 3, perform check by jog operation.)	

2	Greasing to robot drive system	Apply the grease from the grease nipples.	2.3.5 2.3.6
3	Timing belts	Remove each cover and check each timing belt for tension.	2.3.7

d) Annual (every year) inspection

No.	Item	Inspection procedures	Relevant Para.
1	Replacement of position detector batteries (Replace the batteries every year.)	Replace the position detector backup batteries. (They are housed in the robot slide base.) If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, a battery alarm will occur. Replace the batteries immediately then.	2.3.7

e) Overhaul (every five (5) years)

No.	Item	Inspection procedures	Relevant Para.
1	Inspection of consumable parts	Inspect the flexible cables, reduction gears, linear guides, timing belts and motors, and replace them when necessary. Also change the grease of the reduction gears.	
2	Battery replacement (Replace the battery every five (5) years.)	Replace the RAM (controller) backup battery. If the running time is long, battery consumption is small. If the power OFF time is long, the consumption is large. If the battery voltage drops, the alarm message saying "1–145 MAIN Battery alarm" will generate. Replace the battery immediately then.	3.3.3

No.	Item	Inspection procedures	Relevant Para.
3	Replacement of switching power supply units (Replace the units every five (5) years.)	As the switching power supply unit uses an aluminum electrolytic condenser, its life is limited. Replace each switching power supply unit (DC5 V, DC24 V) for preventive maintenance. The life of the switching power supply unit differs with load current, running hours and ambient temperature. If the load current is large, the running time is long and the ambient temperature is high, the life becomes short.	3.3.4

1.3 Maintenance Tools

We recommend to use the following tools for maintaining the robot and controller.

- Phillips screwdrivers
- Hexagonal wrench set: M3 ~ M16
- Assembly wrench set
- Water pump pliers
- Nippers
- Grease guns

1.4 Maintenance Contract and Repair

1.4.1 Maintenance Contract

With the exception of greasing and lubrication, we recommend that the user conclude with us an after-sale service contract on the inspection items for maintenance and inspection to be done semi-annually and after.

1.4.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the Toshiba Machine Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller.

- Robot model
- Manufacture number
- Date of manufacture
- Operating time

1.4.3 Modification

This robot and controller MUST NOT be modified without the consent from Toshiba Machine.

! CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

Section 2 Maintenance of the Robot

2.1 Cautions on Maintenance and Inspection

When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

! DANGER

- Be sure to turn off the main power switch of the controller before approaching the robot for maintenance and inspection.

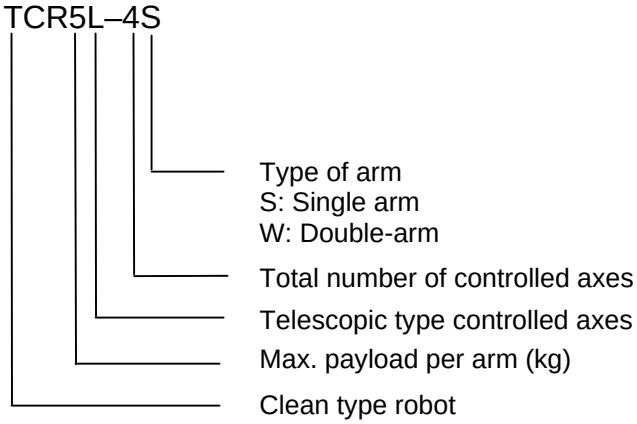
! CAUTION

If the belt tension has been adjusted, the home point of the machine system may shift. When this happens, re-teaching, etc., will be required.

2.1.1 Type of Small-Size Liquid Crystal Transfer Robot

The type of the small-size liquid crystal transfer robot is assigned as shown below, according to the type of the arm (i.e., single arm, double-arm, linear arm) and absence or presence of the traverse shaft.

Example:



2.2 Layout of Robot Components and Drive Mechanism

This paragraph describes each axis drive mechanism.

<Basic structure of TCR5L (double-arm)>

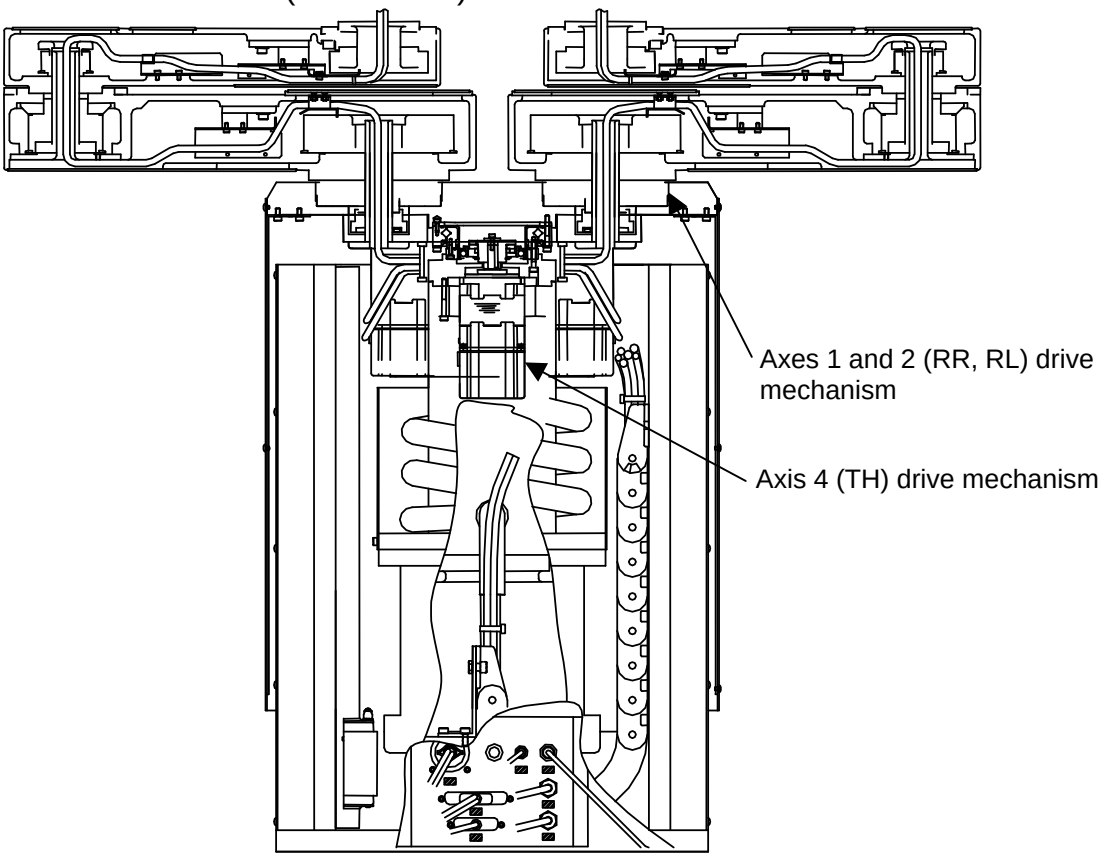


Fig. 2.2 Internal structure

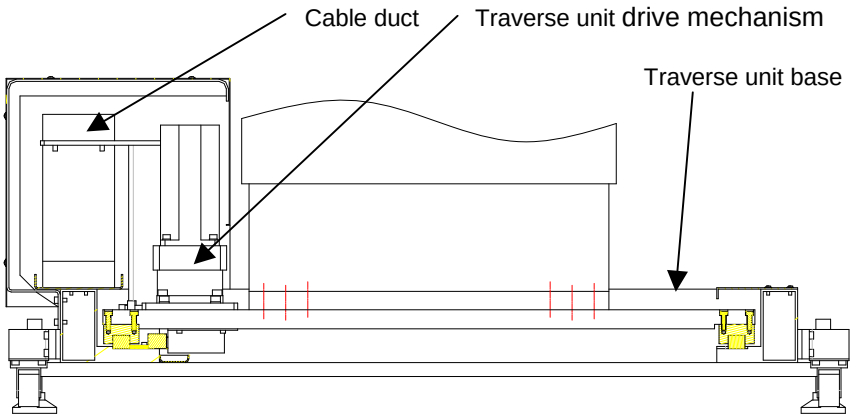


Fig. 2.3 Internal structure of the traverse unit

a) Axes 1 and 2 (RR, RL)

The arms of axes 1 and 2 are extended or retracted. Rotation of the AC servo motor is input to the hollow type harmonic reduction gear via the timing belt. The output of the harmonic reduction gear rotates the arm 1. At the same time, this rotation is input to the arm 2 and tool flange at the end of the arm 2 via the timing belt housed in the arm. As a result, the hand and fork attached to the tool flange are extended or retracted.

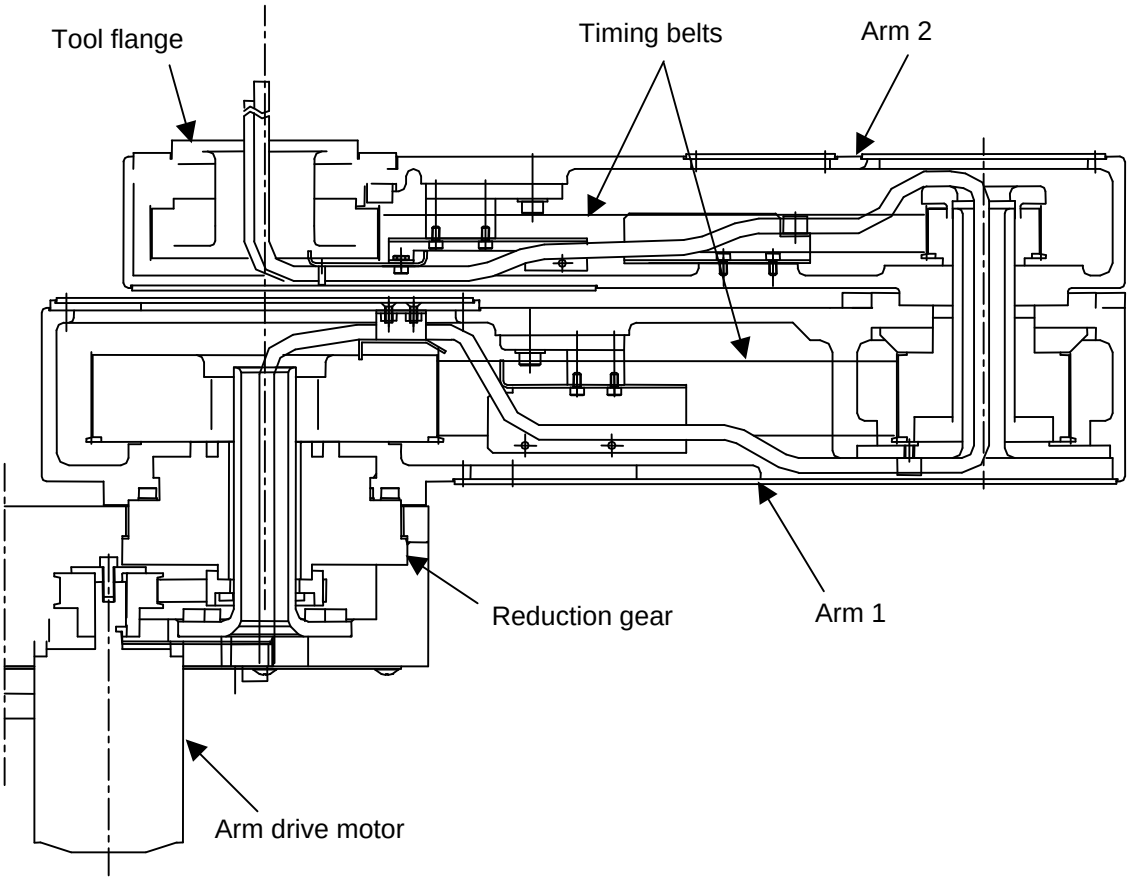


Fig. 2.4 Axes 1 and 2 drive mechanism

b) Axis 3 (Z)

The Z-axis moves up and down the arm.

Rotation of the AC servo motor is input to the ball screws via the timing belt to vertically move the arm. Linear guides are used for the guides.

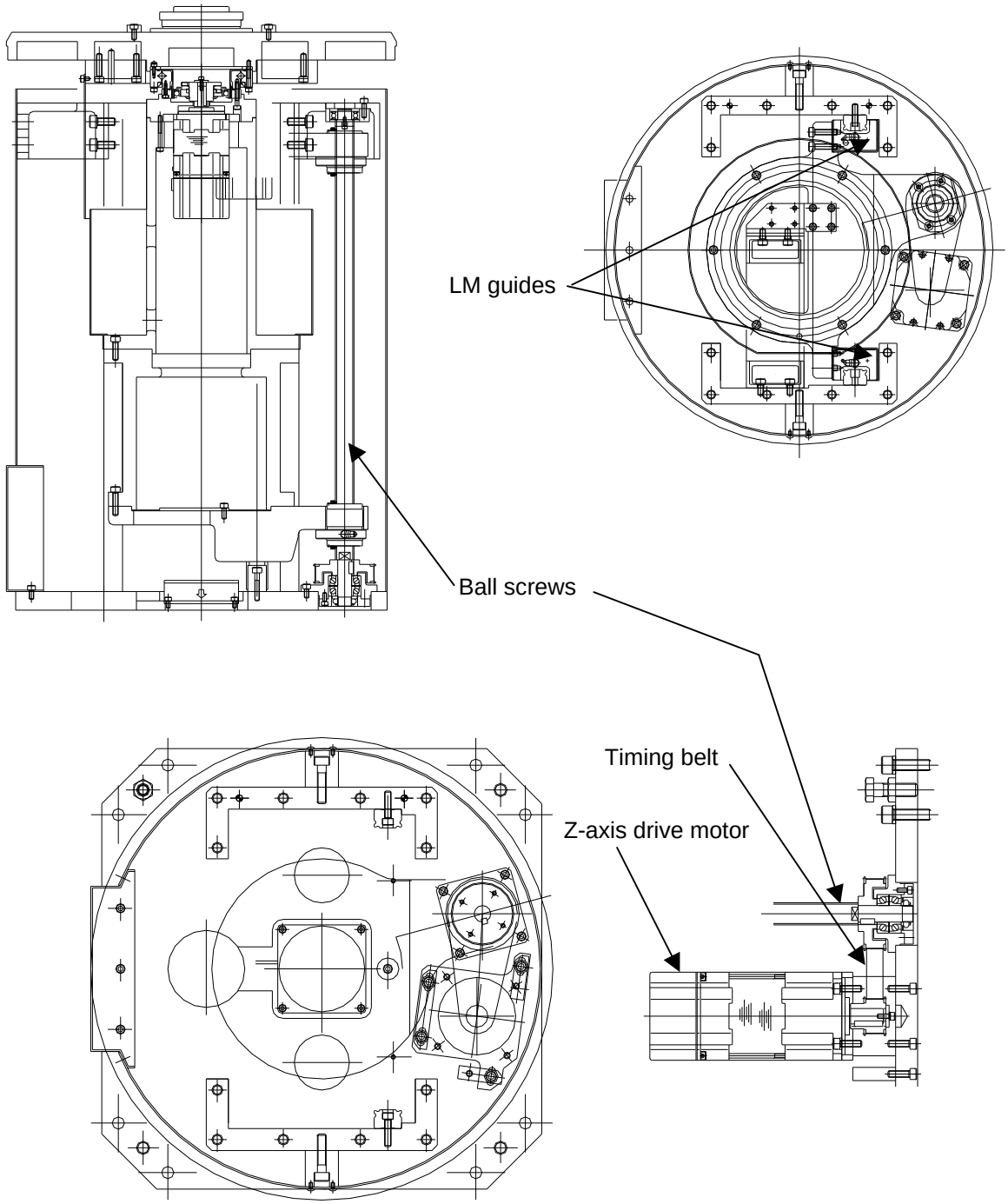


Fig. 2.5 Axis 3 drive mechanism

c) Axis 4 (TH)

The axis 4 (TH) swivels the entire arm.

Rotation of the AC servo motor is input to the harmonic gear via the gears. The output of the harmonic reduction gear swivels the entire arm.

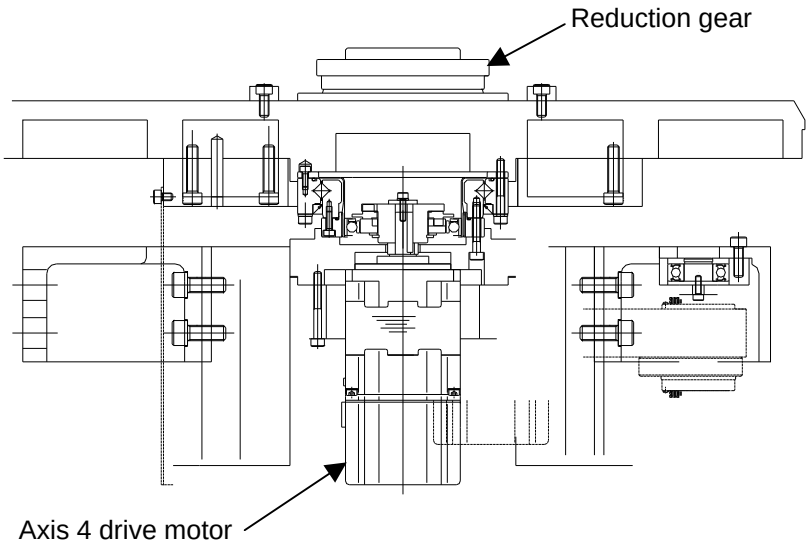


Fig. 2.6 Axis 4 drive mechanism

d) Axis 5 (Traverse Axis)

The axis 5 has the function for moving the robot body.

The output of the AC servo motor is transferred to rack & pinion via Reduction gear to move the robot body on the slide base.

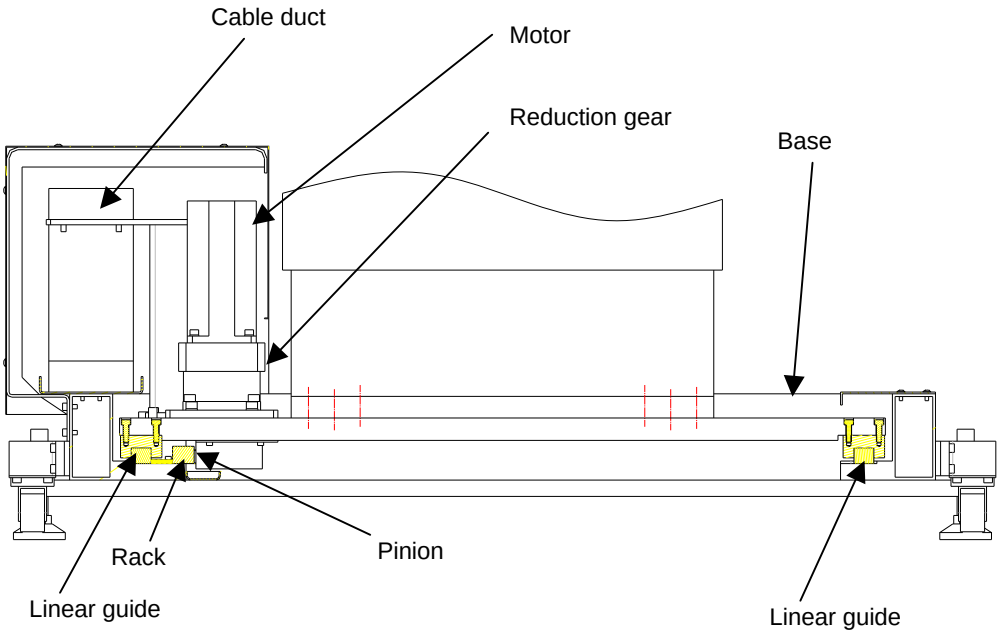


Fig. 2.7 Axis 5 drive mechanism

2.3 Robot Maintenance Procedures

2.3.1 Check of Cable and Duct for Abnormality

a) Before starting the robot operation, make sure that each cable and duct are not twisted, bent or cracked locally. If any abnormality is found, replace the cable or duct with a new one to prevent cable breakage, etc.

b) Each connector securing the duct is clamped and secured with a steel plate and hexagon nut. If the hexagon nut is loosened, tighten it, using water pump pliers, etc.

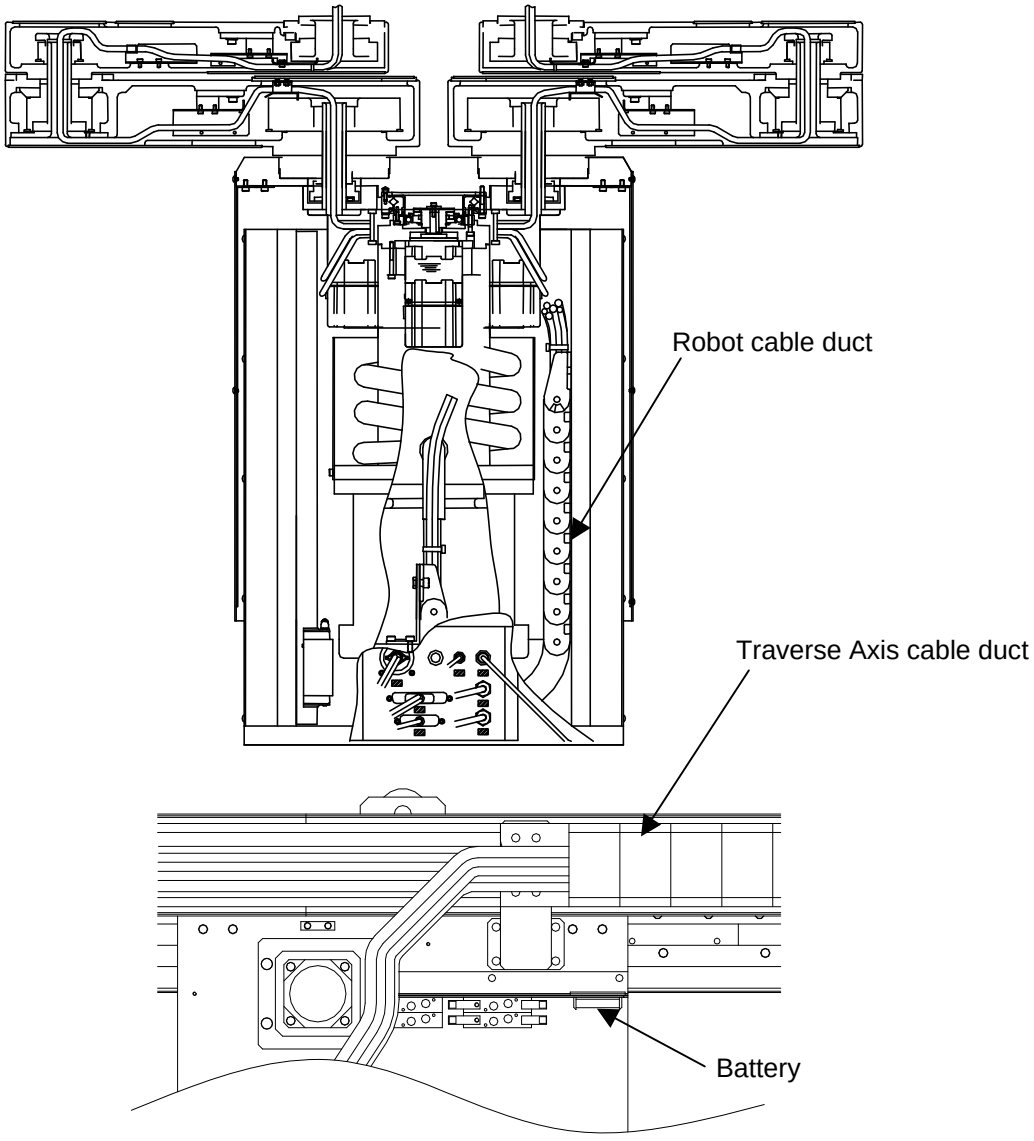


Fig. 2.8 Cable duct secured condition

2.3.2 Check of Set/Clamp Screws and Bolts of Each Part for Tightening

Before the operation, make sure of the following items visually and by touching with the hand.

a) Make sure that the bolts securing the robot body and traverse shaft frame are tightened completely.

b) Make sure that the clamp screws of each cover are tightened completely.

2.3.3 Check of Set/Clamp Screws and Bolts of Each Part for Complete Tightening

Completely tighten the bolts and screws in Para. 2.3.2 above, then make sure of them again.

2.3.4 Check of Each Cable for Wear

Remove the cover of each part and make sure that each cable is not worn out, cut or scratched, or stained with contaminant. If any abnormality is found, the cable should be replaced with a new one.

2.3.5 Check of Rack and Pinion for Lubricated State, and Greasing

Make sure that the tooth surface of the rack is lubricated properly. If it is dry, apply the grease to the tooth surface of the rack with hand.

Fill the grease to the linear guide from the four (4) grease nipples of the guide block until it gushes over the block exterior, using a grease gun.

After the greasing, move the axis 5 over the full stroke a few times and wipe out any surplus oil.

Grease used: AFE Grease (THK)

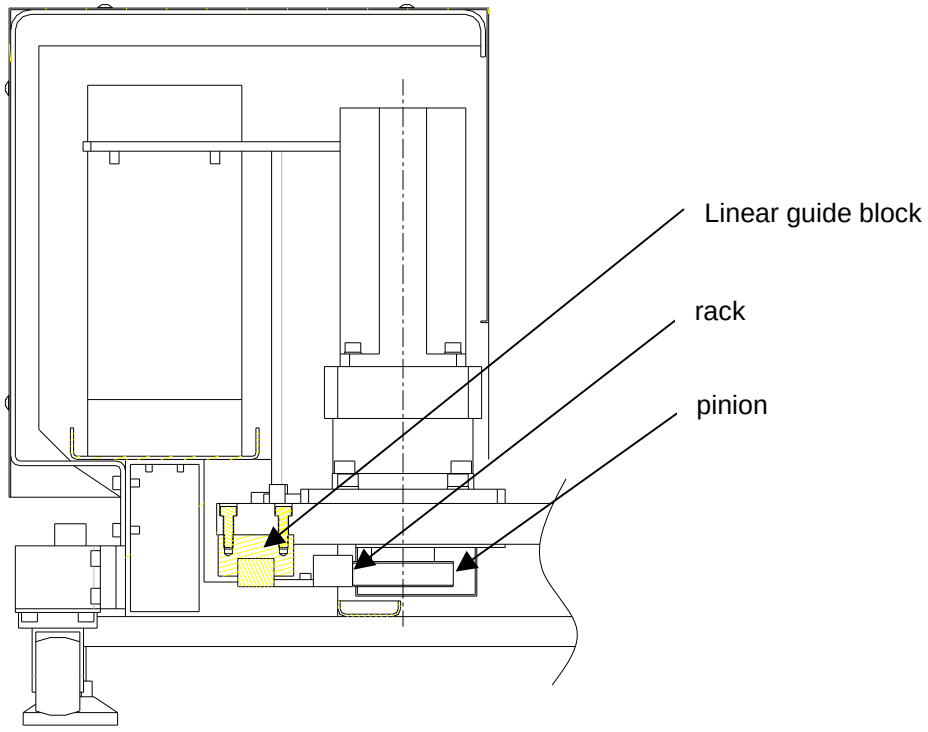


Fig. 2.9 Greasing to the traverse unit

2.3.6 Greasing to Robot Drive System

Check of Rack and Pinion for Lubricated State, and Greasing grease (made by THK).

To grease the LM guides, and arm drive gears, remove the cover, and you can see the greasing nipples there. Fill the AFE grease (white-colored, made by THK) from these nipples until the grease gushes over, using a grease gun. After the greasing, move the axis 5 over the full stroke a few times and wipe out any surplus oil.

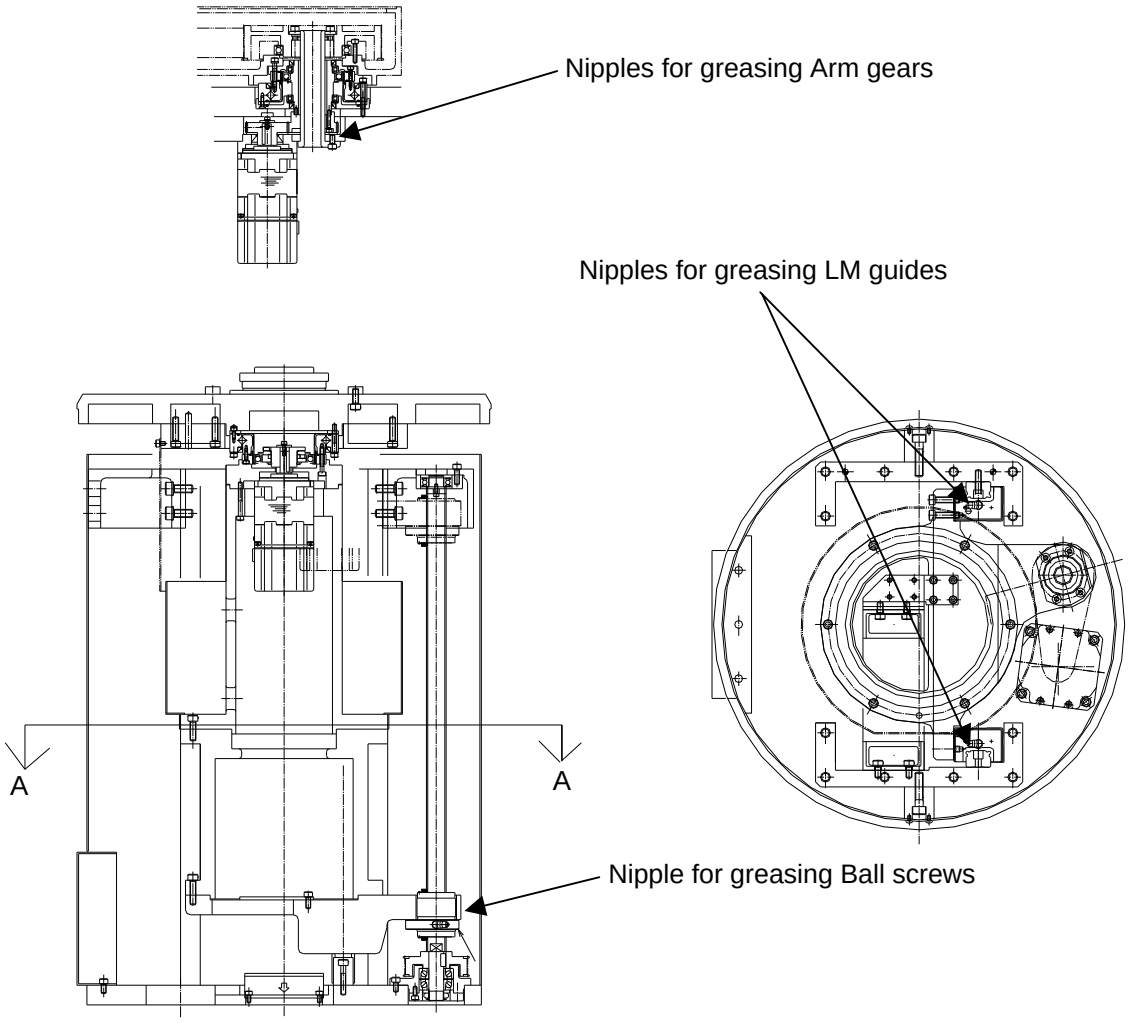


Fig. 2.10 Greasing to robot drive system

! CAUTION

Take careful precautions not to mistake the type of the grease you use.
Otherwise, the robot performance will drop and the robot will go wrong.

2.3.7 Check for Timing Belt

This robot uses timing belts to extend and retract, and move up and down the arm. Tension of the timing belts is factory-adjusted after they are operated continuously and sufficiently to eliminate the initial belt elongation.
The belt will elongate, however, with the lapse of the operating time and regular check is required. Push each belt by hand and make sure that it has appropriate tension.

Unless the tension is enough, the robot position may shift.
Under the following occasions, the timing belt should be replaced with a new one.

[1] When a crack has developed on the root or rear side of the belt tooth.
[2] When the belt has swollen due to oil left on the belt.
[3] When the belt tooth has worn out (by about half the tooth width).
[4] When the belt tooth has jumped due to the worn belt tooth.

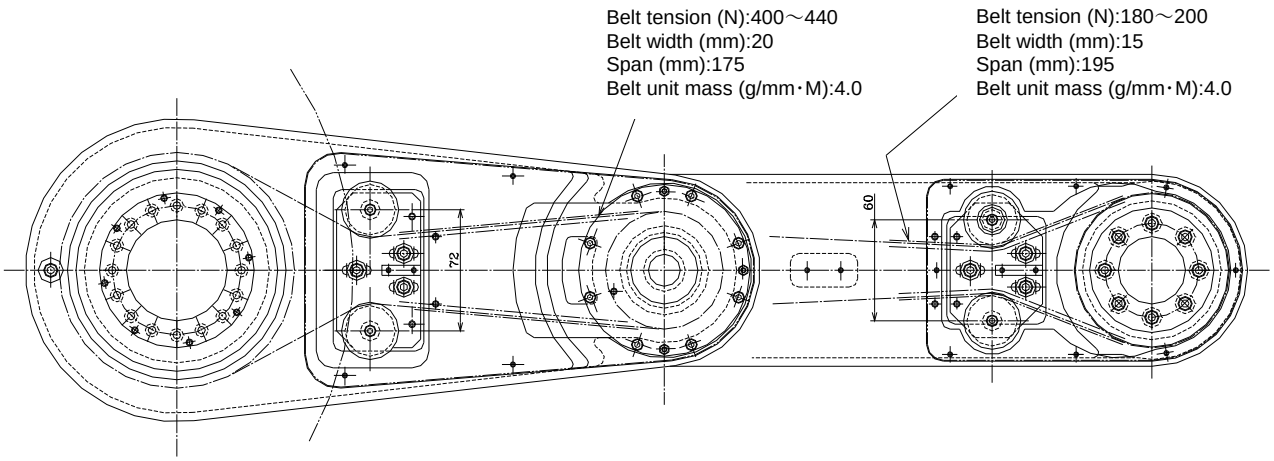


Fig. 2.11 Arm drive belt

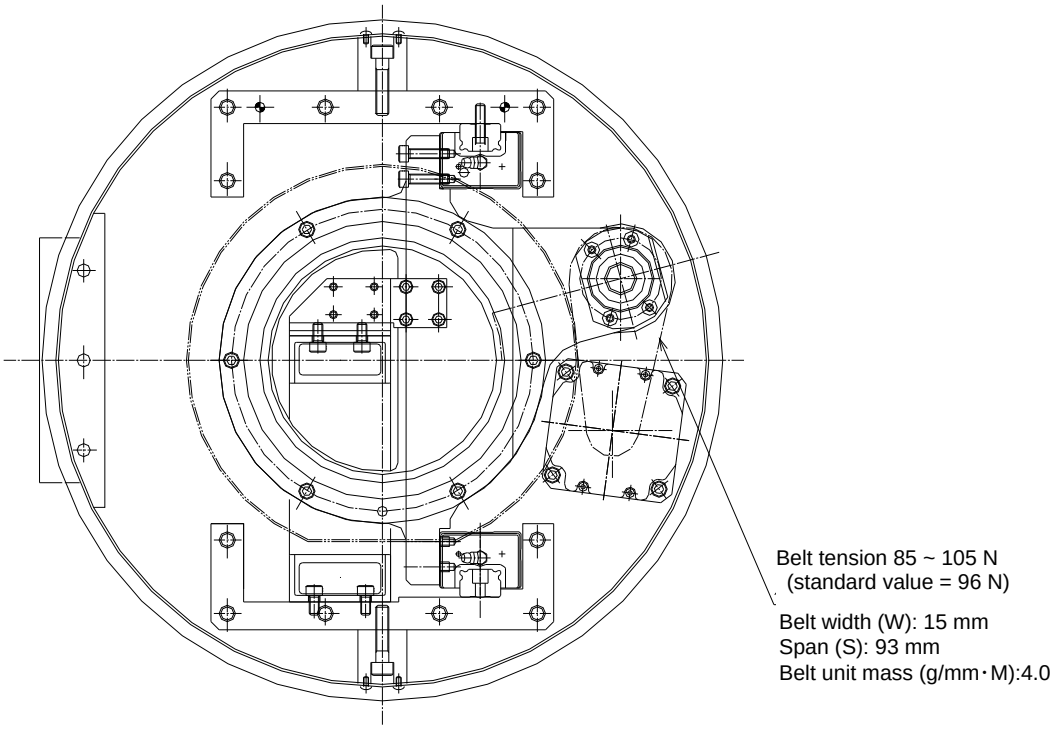


Fig. 2.12 Vertical mechanism drive belt

2.3.8 Replacement of Position Detector (or Encoder) Batteries

To back up the data of the position detector attached to each motor, batteries (alkaline dry batteries) are used.
Though the battery life will vary with the controller power OFF time, be sure to replace the batteries with new ones at the time of annual inspection. When the robot is not used for a long term (i.e., the batteries are left intact in the power OFF condition), replace the batteries at the time of robot startup. As a yardstick, the shutdown period is two (2) months.
The battery replacement procedures are given below.

<Replacement procedures>

1) Keep safety and turn the power on. Press and set ON the EMERGENCY pushbutton switch to stop the robot. If the safety cannot be kept, turn the power off once.
2) The battery box (Fig. 2.13) is secured to panel in the traverse base. Remove the battery box. NEVER disconnect the wiring.
3) Remove the battery box cover, then the batteries.
4) Set the new batteries (size AA battery × 3 pcs.), taking careful precautions not to mistake the "+" and "–" polarities. When you wish to replace the batteries while the power is turned off, start replacing the batteries after lapse of one (1) minute or over because the power failure operating current runs for several tens seconds, then finish the replacement within five (5) minutes.
5) Secure the battery box to the panel again.
6) If a battery alarm has generated, reset it through the teach pendant.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

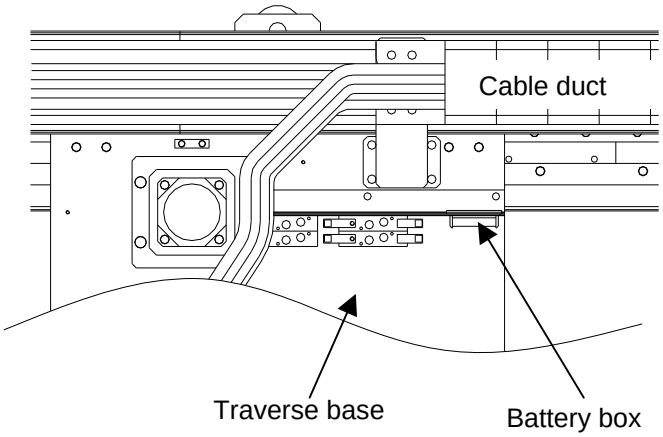


Fig. 2.13 Battery set position (with the traverse axis)

Unless the traverse axis is provided, the battery box is attached to the robot body.

Cautions:

If the battery voltage has dropped, either of the following battery alarms will occur.

1–049 Axis1 Battery alarm
1–065 Axis2 Battery alarm
1–081 Axis3 Battery alarm
1–097 Axis4 Battery alarm
1–113 Axis5 Battery alarm

When the batteries have been replaced just after the battery alarm generation, the battery voltage returns to normal and relevant battery alarm is reset automatically. Otherwise, the battery voltage drops further with either of the following position detection error generated.

8–065 Axis1 Encoder abnormal
8–097 Axis2 Encoder abnormal
8–129 Axis3 Encoder abnormal
8–161 Axis4 Encoder abnormal
8–193 Axis5 Encoder abnormal

The above errors signify that the position data detected by each encoder is not reliable any further. When this happens, the robot enters an emergency stop state to prevent the servo from turning on. Also, if the power is turned off in this condition, the position data is lost.

If the “axis encoder abnormal” error has generated, restore the system, referring to Para. 2.3.9 below.

2.3.9 Restoring Home Point (or Origin) after Home Point Data Is Lost

If the position encoder batteries have dropped and remain intact, the position data of the home point will be lost.
This paragraph deals with the home point setting procedures after the home point data has been lost.
The position encoder batteries are used to save multi-turn data of the encoder. When you align each axis with the home point match-mark visually and reset the multi-turn data, you can restore the position data.

<How to restore the home positions of axis 1, 2,4, 5>

[1] Turn off the servo power, and manually align an axis whose home point you wish to set with the home point match-mark.

[2] Select the TEACHING mode by means of the master mode switch, then press the “UTILITY” key equipped on the teach pendant.

[3] When the [UTILITY] screen is called, press the [NEXT] key once, then press the [F5] function key corresponding to the [ENC] menu.

[4] When the encoder information screen appears, press the “ALT” and “0” keys at the same time to cancel the editing lock mode.

[5] Set the cursor to the axis whose multi-turn data you wish to reset to zero (0), and press the [F3] function key corresponding to the [RESET] menu.

[6] Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

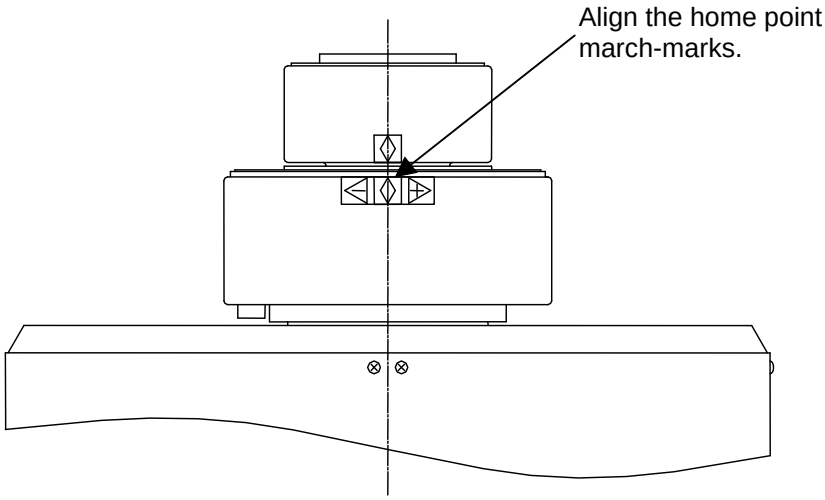


Fig. 2.14 Axes 1, 2 home positions

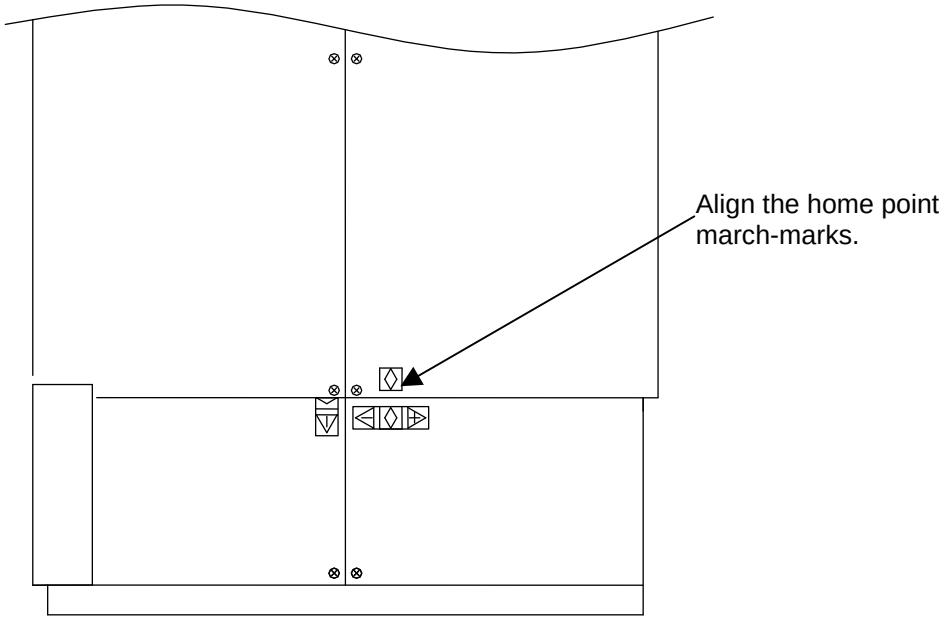


Fig. 2.15 Axes 4 home positions

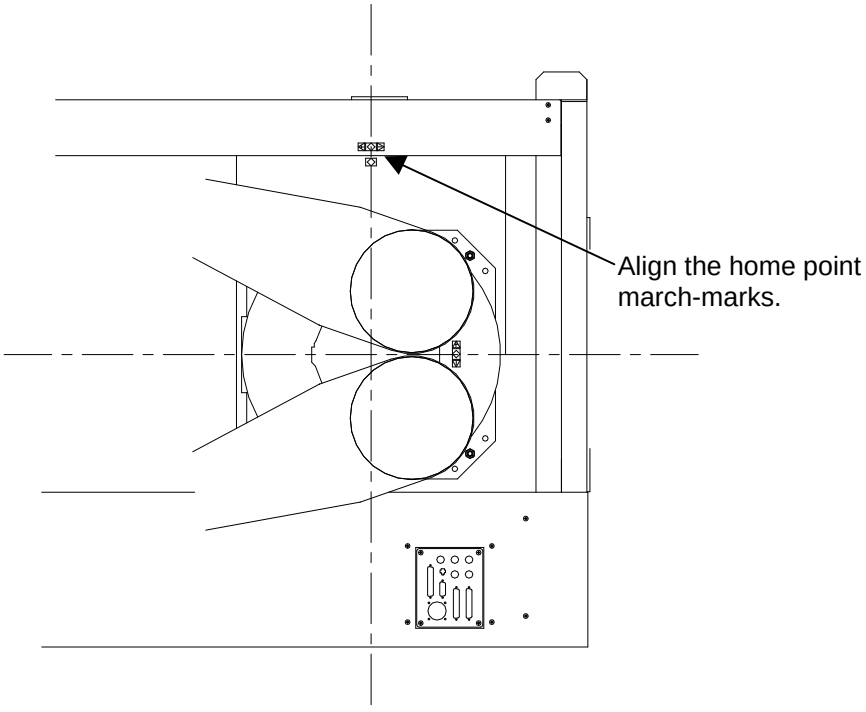


Fig. 2.16 Axes 5 home positions

<How to restore the home position of axis 3>

As brake is effected on the axis 3 motor, it is not possible to manually align it with the home point match-mark. You can restore the position data, however, when you set the current position as the temporary home point, align the axis with the home point match-mark by jog operation, and manually enter the encoder offset default data registered beforehand into the system file (MACHINE.PAR).

a) Record the encoder offset default data on the clean paper, etc.

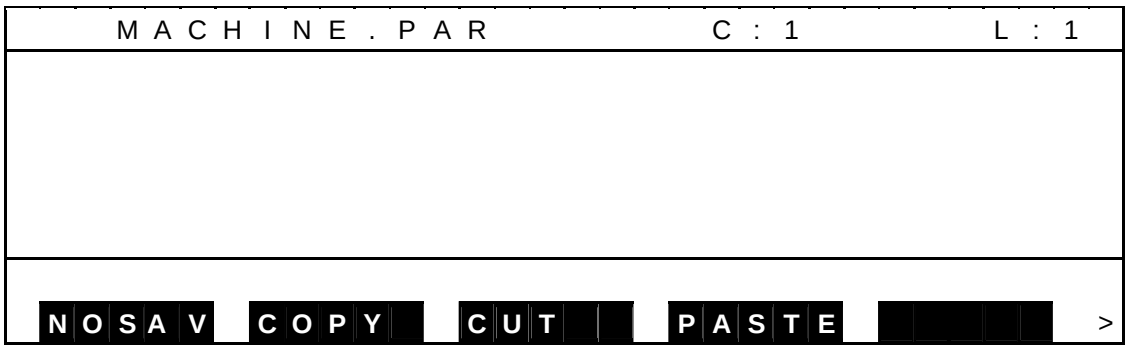
1) While the servo power is turned off, select the TEACHING mode by means of the master mode switch, then open the machine parameter file ("MACHINE.PAR") by selecting the "EDIT" key provided on the teach pendant. The "MACHINE PAR" file contains the set data characteristic of the machine, such as home points and options.

```
  FILE SELECT                                  (001/256)
* MACHINE.PAR_                    6   Feb-12-2001
  FILE0002.012  123456   Feb-13-2001
  FILE0003.012  123456   Feb-14-2001
  FILE0004.012  123456   Feb-15-2001
  FILE0005.012  123456   Feb-16-2001
> _
  NAME   TIME   NEW
```

2) Refer to the data under "[M05] Servo offset", as shown below. The third (3rd) value starting from the left (i.e., "–164946" in the example below) signifies the axis 3 encoder offset value. Record it on the clean paper, etc. When you set the temporary home point later, the value is updated. Be sure to record the value.

```
[M05] Servo offset
= 468268   -956224   -164946   636319   332680   0
```

3) Select [NOSAV] and close the "MACHINE.PAR" file without saving the data.



b) Set the current position as the temporary home point.

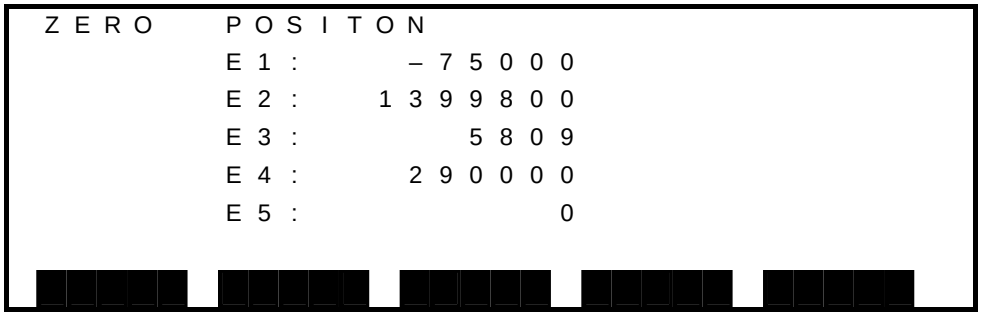
1) Select the TEACHING mode by means of the master mode switch.

2) Press the “UTILITY” key provided on the teach pendant.

3) Press the [NEXT] key (F6) twice on the teach pendant.

4) Press the [F3] key on the teach pendant to select the ZEROP mode.

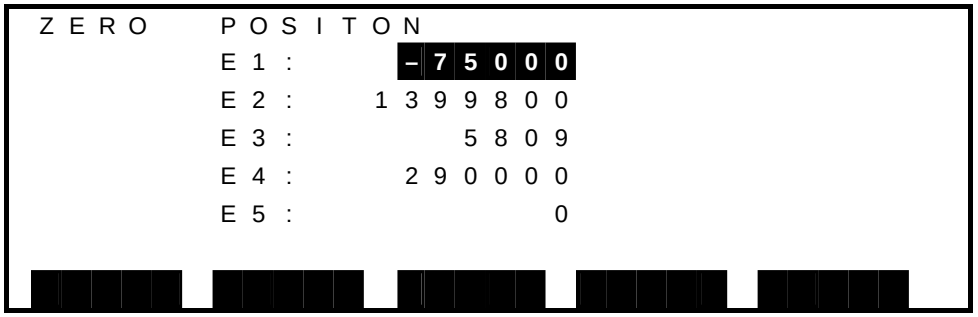
5) The following home point setting screen appears on the display.



6) Press the “ALT” and “0” keys on the teach pendant at the same time.

7) The following home point editing screen is displayed.

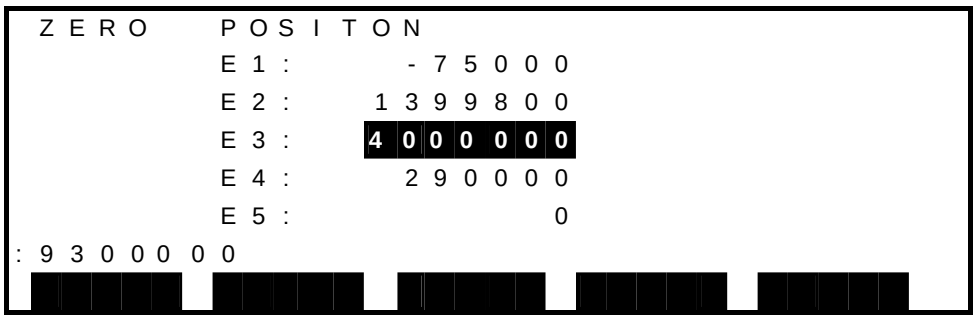
8) The value of E1 (axis 1) is indicated by reversed image.



9) Press the "↓" key on the teach pendant twice. Then the value of E3 (axis 3) is indicated by reversed image.

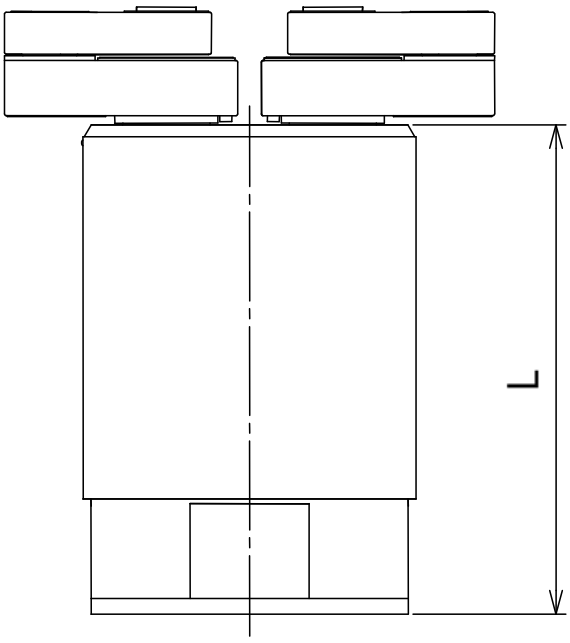
10) Press the "EXE" key provided on the teach pendant, and enter "**000000" in E3. For **, specify a two (2)-digit number indicating robot type Z**.

Example: When the robot type is Z stroke 400mm, enter the data as shown below.



11) After the data has been entered, return to the previous screen by pressing the "ESC" key.

c) Turn the servo power on and slowly move down the axis 3 to align it with the home point match-mark or the position of Fig 2.17.

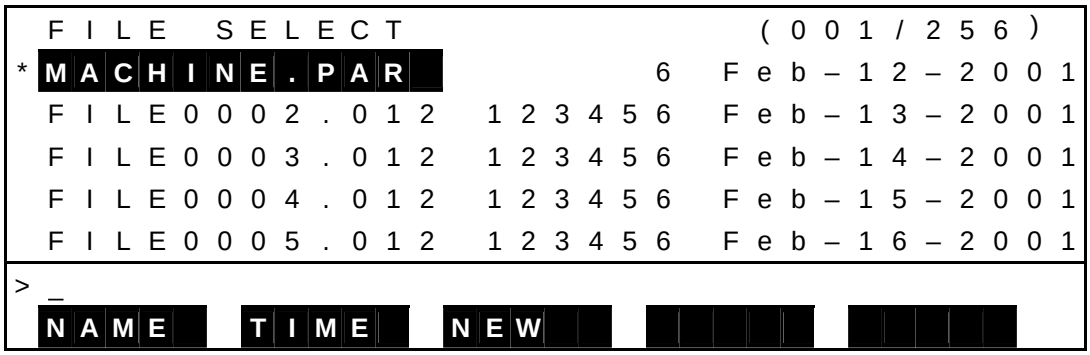


L=620(mm)

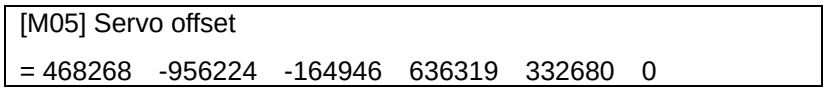
Fig. 2.17 Axes 3 home positions

d) Enter the encoder offset default value.

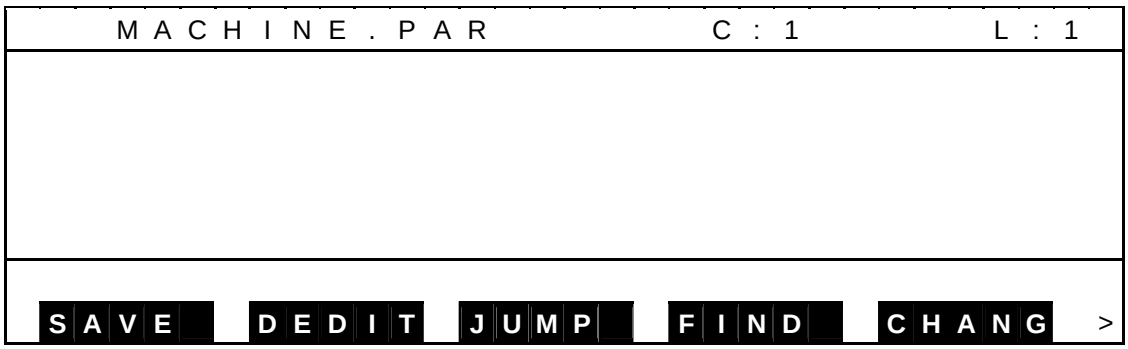
1) While the servo power is turned off, select the TEACHING mode by means of the master mode switch, then open the machine parameter file ("MACHINE.PAR") again by selecting the "EDIT" key provided on the teach pendant.



2) Change the value registered under "[M05] Servo offset", as shown below. The third (3rd) value starting from the left (i.e., "–164946" in the example below) signifies the axis 3 encoder offset value. Replace this value with the value already recorded in Para. a) above then.



3) Select [SAVE] to save and close the "MACHINE.PAR" file.



e) Turn the controller power off once, then on again. Finally, move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

Section 3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of robot controller, follow the items given below so that the work can be carried out safely.

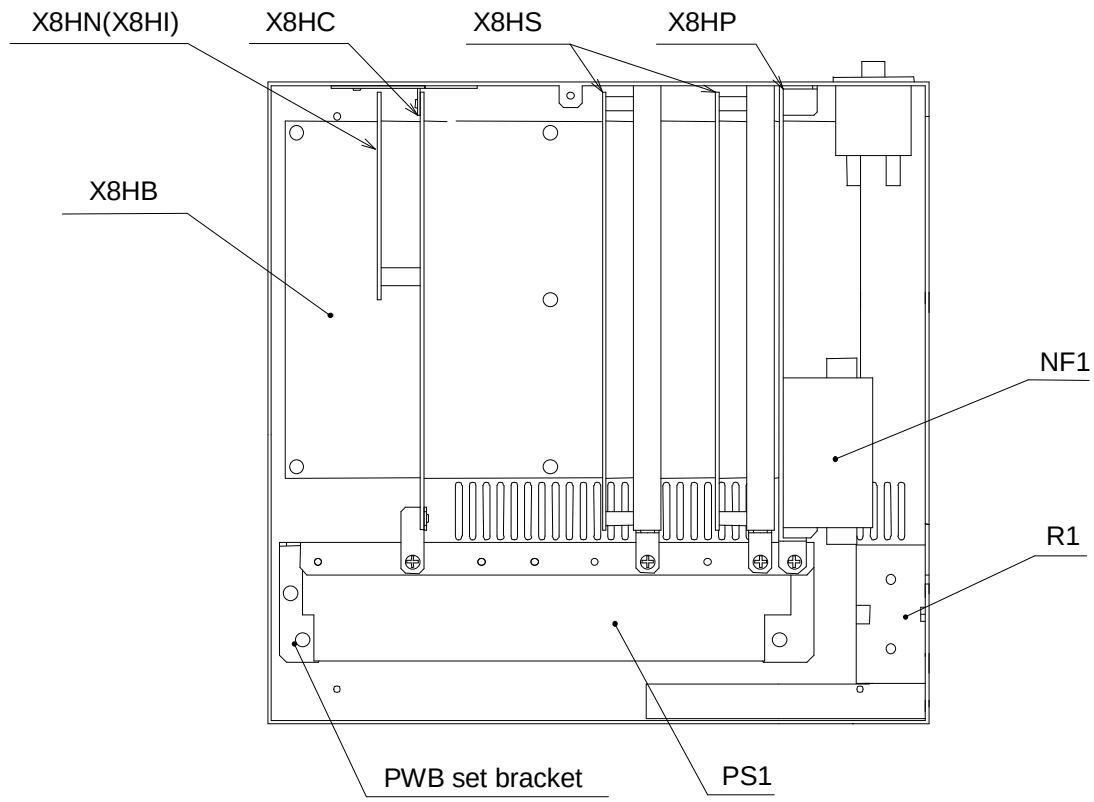
! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts



Upper side view

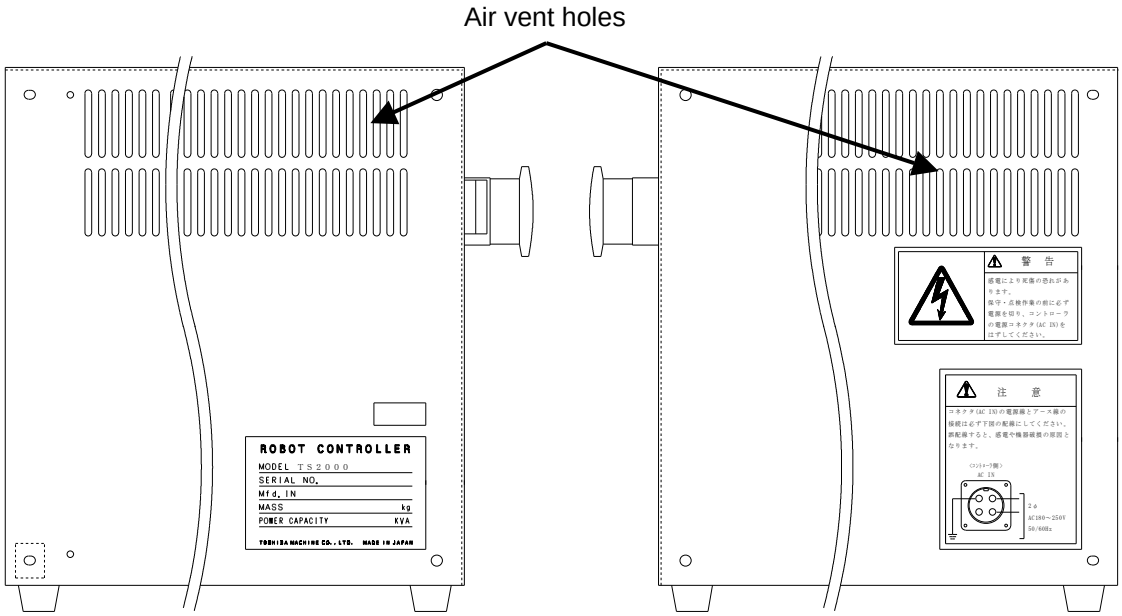
Fig. 3.1 Layout of controller parts

Part name	Descriptions
PS1	P5V, P24V switching power supply
PWB set bracket	PS1 set bracket
X8HP	Servo power printed board
X8HC	Main control printed board
X8HN (X8HI)	I/O output printed board
X8HS	Servo printed board
X8HB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.3 Maintenance Procedures

3.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction. To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.



Left side view Right side view

Fig. 3.2 Controller side views

a) Make sure that no obstacle stands in front of the air vent holes. If there is any obstacles, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.3.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

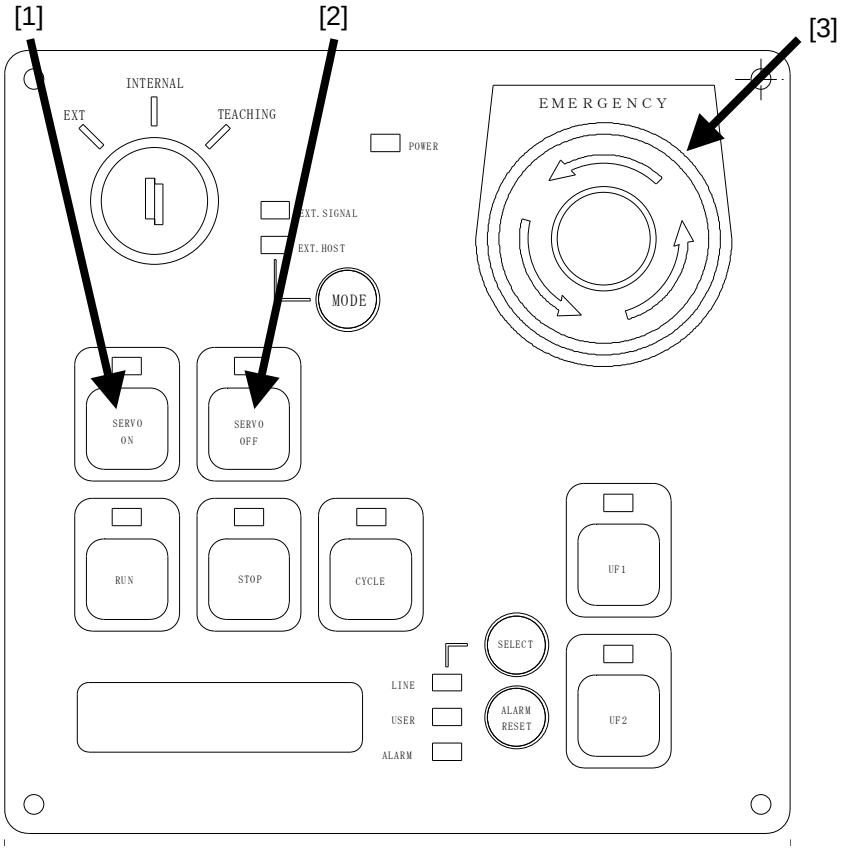


Fig. 3.3 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [2] provided on the control panel functions properly.

1) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

2) Press the EMERGENCY switch [2] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [3] LED is illuminated. At this time, make sure that the EMERGENCY switch [2] remains depressed.

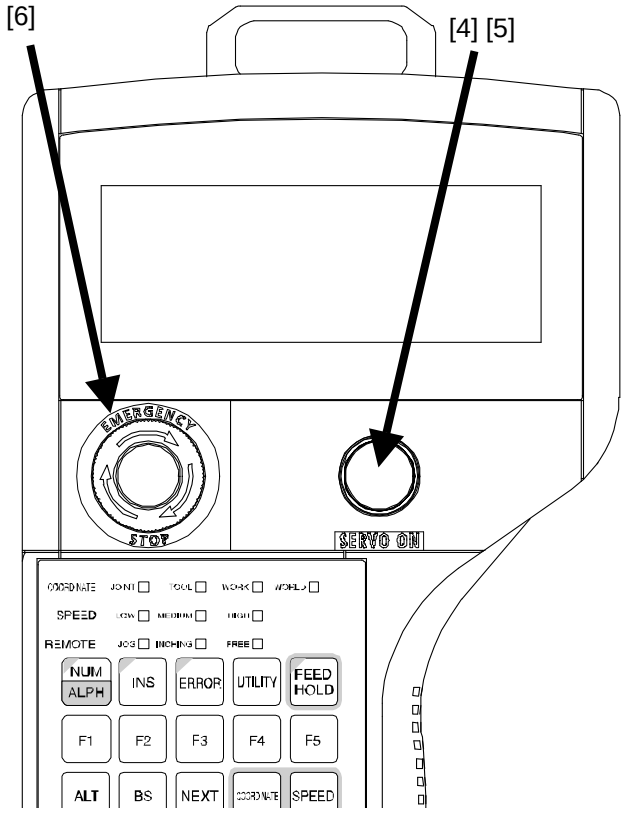
3) Turn the EMERGENCY switch [2] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Press the SERVO ON switch [4] on the teach pendant to turn on the servo power, and make sure that the SERVO ON switch LED [5] is illuminated.

Note: The servo power can be turned on only in the TEST RUN mode.

2) Press the EMERGENCY STOP switch [6] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out.



At this time, make sure that the EMERGENCY STOP switch [6] remains depressed.

3) Turn the EMERGENCY STOP switch [6] clockwise and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.3.3 Battery Replacement

SRAM equipped on the X8HC printed board of the TS2000 robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the SRAM data, resulting in data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

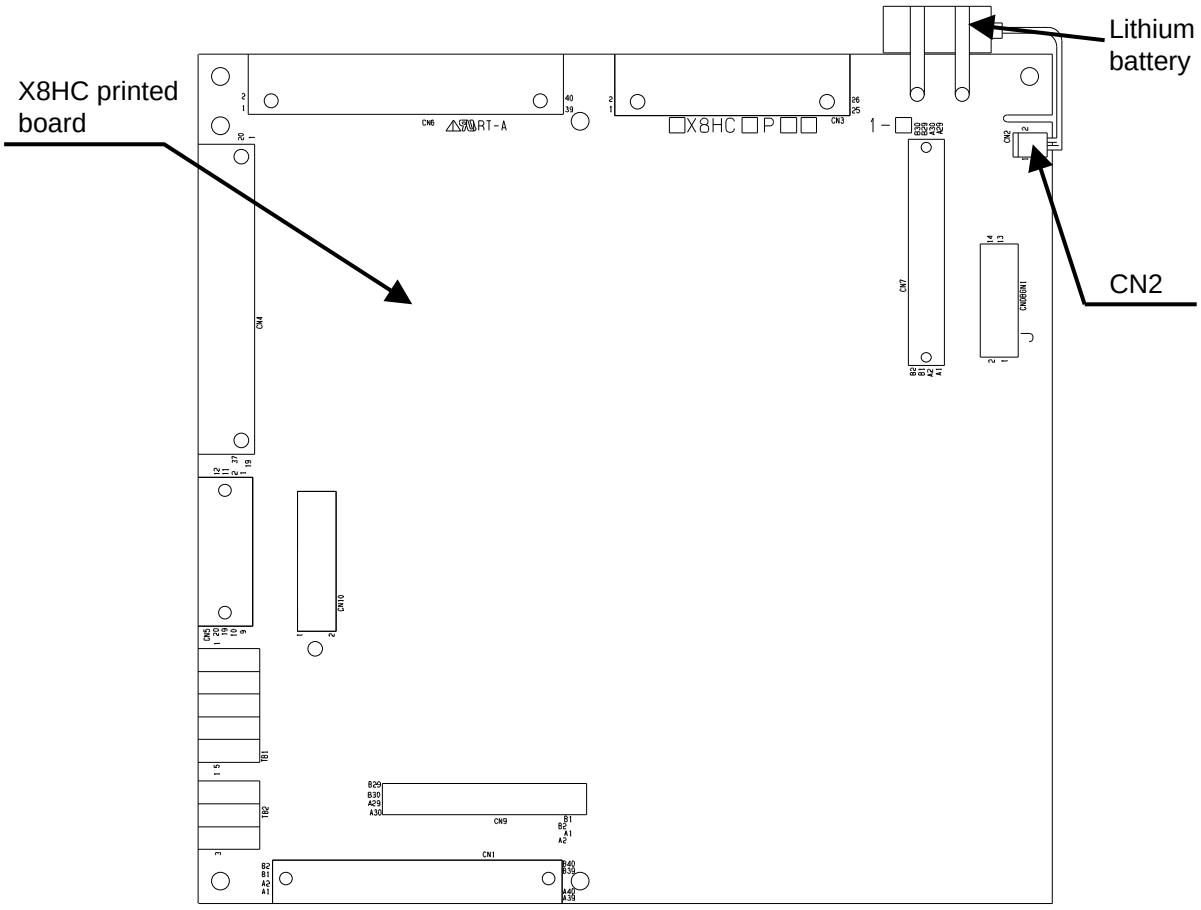


Fig. 3.5 Location of lithium battery

Table 3.1 Type of battery

Type of battery	Manufacturer	Battery service life
ER3V－C	Toshiba Denchi	10 years (at room temperature)

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.

Only the battery shown in Table 3.1 should be used. As this is an exclusive battery, contact Toshiba Machine at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than five (5) minutes, copy all programs stored in the SRAM to the personal computer. In the worst case, all memory data are lost.

2) Set OFF the main power switch.

3) Remove the cover from the body.

4) Remove the battery connector from CN2 of the X8HC printed board, then disconnect the battery from the Tie-wrap securing the battery.
(For the battery position, see Fig. 3.5.)

5) Set a new battery within five (5) minutes.
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Mount the cover on the body.

7) If the battery is kept removed for more than five (5) minutes, turn on the main power switch and load the programs from the floppy disks.

If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Waste battery should be disposed of according to the user's in-house regulations.
- NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

3.3.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5V/DC24V) used in the TS2000 controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Remove the cover from the body.

4) Remove all connectors in the control panel, which are linked to the X8HC, X8HS (2 pcs.) and X8HP printed boards. (For the position of each printed board, see Fig. 3.1.)

5) Remove the X8HC, X8HN (X8HI), X8HS(2 pcs.) and X8HP printed boards from the body.

6) Disconnect the cable connecting the power module and the cable between PS1 and mother board.

7) Remove the screws (3 places) which secure the body and PWB set bracket. (For the position of the PWB set bracket, see Fig. 3.1.)

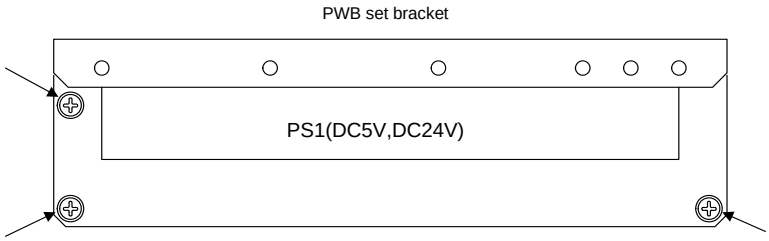


Fig. 3.6 Upper side view

8) Remove the connectors (2 places) linked to the PS1 with the PWB set bracket inclined slightly, then remove the PWB set bracket from the body.

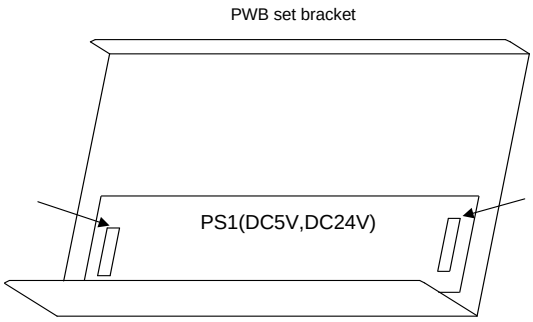


Fig. 3.7

9) Remove each power supply from the PWB set bracket, and set a new power supply.

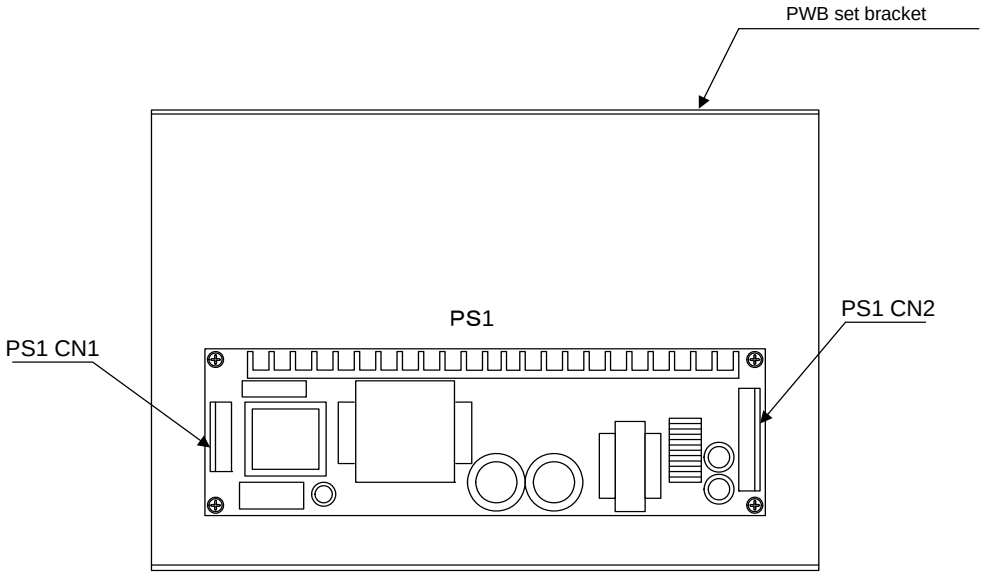


Fig. 3.8

10) Restore the PWB set bracket to the body, where the new power supply is set. Then link the connector.

11) Screw the PWB set bracket into the body (3 places; See Fig. 3.6.)

12) Mount the X8HC, X8HN (X8HI), X8HS (2 pcs.) and X8HP printed boards on the body.

13) Mount the removed connectors on respective printed boards.

14) Mount the cover on the body.

3.3.5 Replacement of Fuse (X8HN, H8HI Printed Boards)

If the current exceeding the specified current has run through the I/O unit, the fuse of the X8HN (X8HI) printed board is blown out. If the alarm saying "I/O FUSE Broken (8–273)" is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

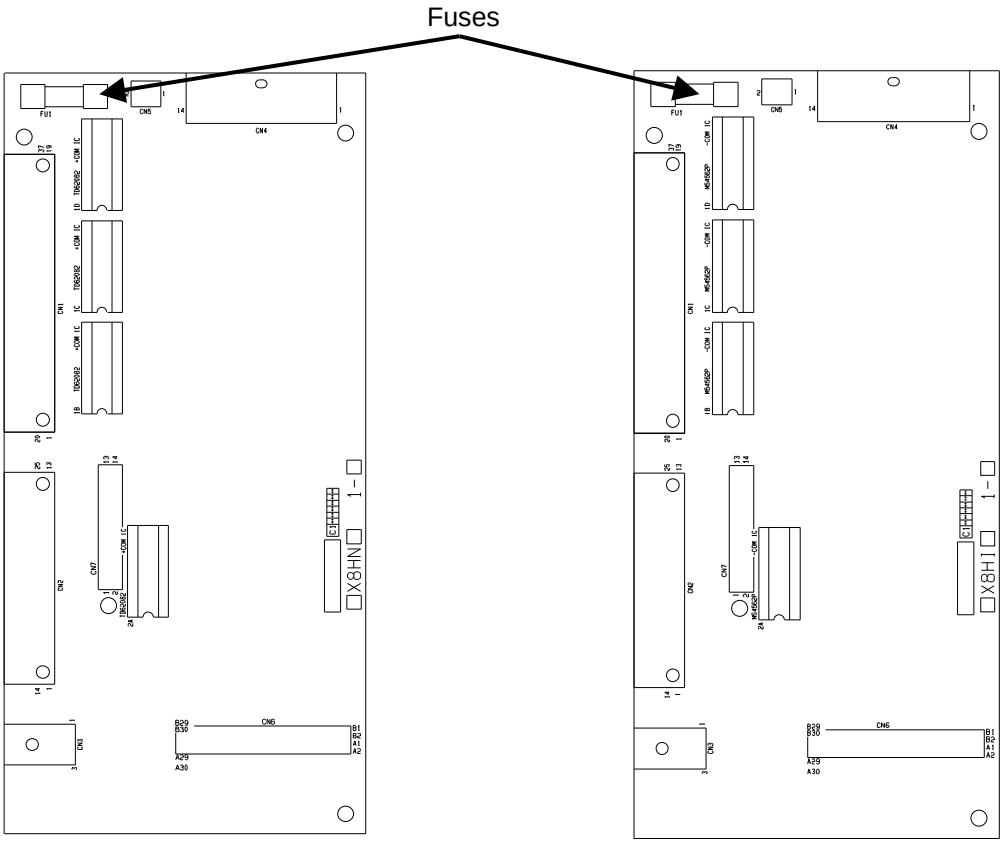


Table 3.1a Type of fuse

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover attached to the fuse unit as shown above.

3) Remove the fuse and set a new fuse.

4) Mount the removed cover.

5) Turn on the controller main power and make sure that no error will occur.

3.3.6 Replacement of Output ICs (X8HN, H8HI Printed Boards)

If the current exceeding the specified current has run through the output unit, the outputs may not turn on normally.
When this happens, the ICs of the X8HN (X8HI) printed board have possibly been damaged. Replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

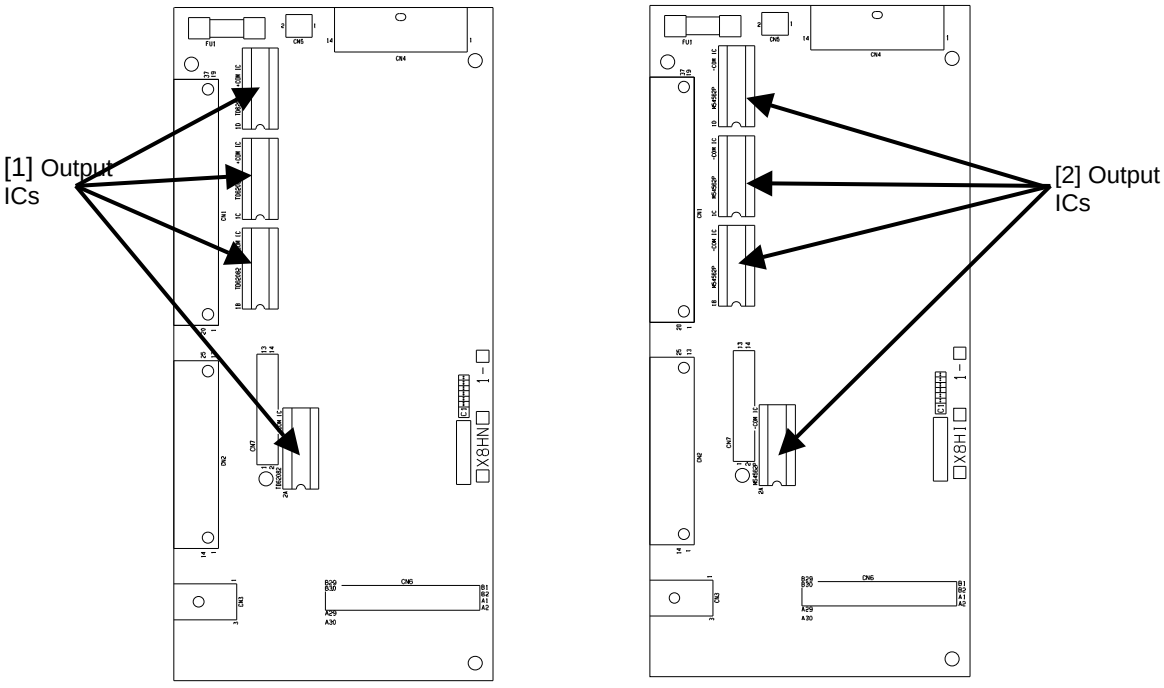


Table 3.1b Type of IC

	Name of board	Type of IC	Manufacturer
[1]	X8HN	TD62082AP	Toshiba Corp.
[2]	X8HI	MM5456P	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the cover from the body.

3) Remove all connectors in the control panel, which are linked to the X8HC printed board.

4) Remove the X8HC and X8HN (X8HI) printed boards from the body.

5) Remove the ICs shown above from respective sockets.

6) Attach new ICs to the sockets.

7) Mount the X8HC and X8HN (X8HI) printed boards on the body.

8) Link the removed connectors to the X8HC printed board, and set the main cover.

9) Turn on the controller main power.

3.3.7 Cautions on Transport

When the robot power cable has been disconnected from the controller, attach the following short-circuit connector (accessory) to the robot power cable.
If the robot has been moved suddenly or heavy vibration has been imposed on the robot at power OFF, without connecting the short-circuit connector, the absolute coordinate counter of the encoder cannot trace any further, resulting in generation of a position sensor error or loss of home position data.

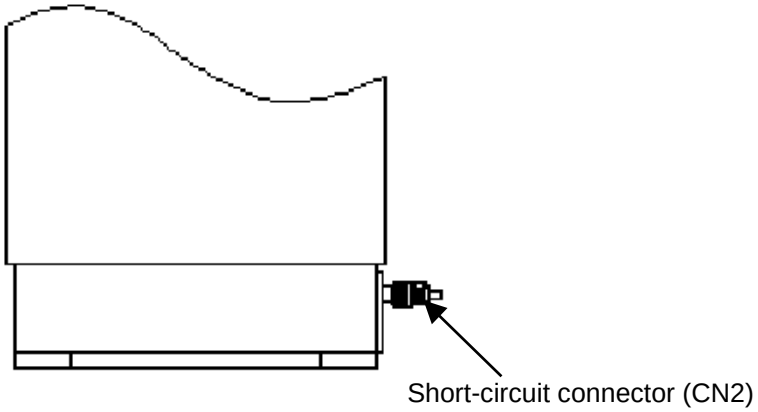


Fig. 3.9 Short-circuit connector set position

Section 4 Home Point Setting

The controller uses absolute encoders. Once the home point has been established, therefore, it need not be set again normally. If the arm belt tension has been readjusted, the motor has been replaced with a new one or the battery error has occurred, however, the home point setting is required again. When this happens, observe the following procedures to establish each home point. It should be noted that the home point setting procedures given below are those which are factory-executed before shipment, and will not guarantee the repeatability of the home point. Therefore, you should reconfirm the teach points, etc.

4.1 Procedures for Setting Each Axis Home Point

(1) Mechanical home point setting of axes 1 and 2 (R1, R2)
The machine home point of the R-axis is set, using simple jigs, as shown below. They are the Jig post (H846568) and the home point jig (M249574).
Mount respective home point jigs in the following manner, then perform the home point setting operation (ZEROP) at this point. And turn on the servo power, advance the arm to the point “-290mm” by jog operation. At the position in which the arm was folded, perform the home point setting operation (ZEROP) again.

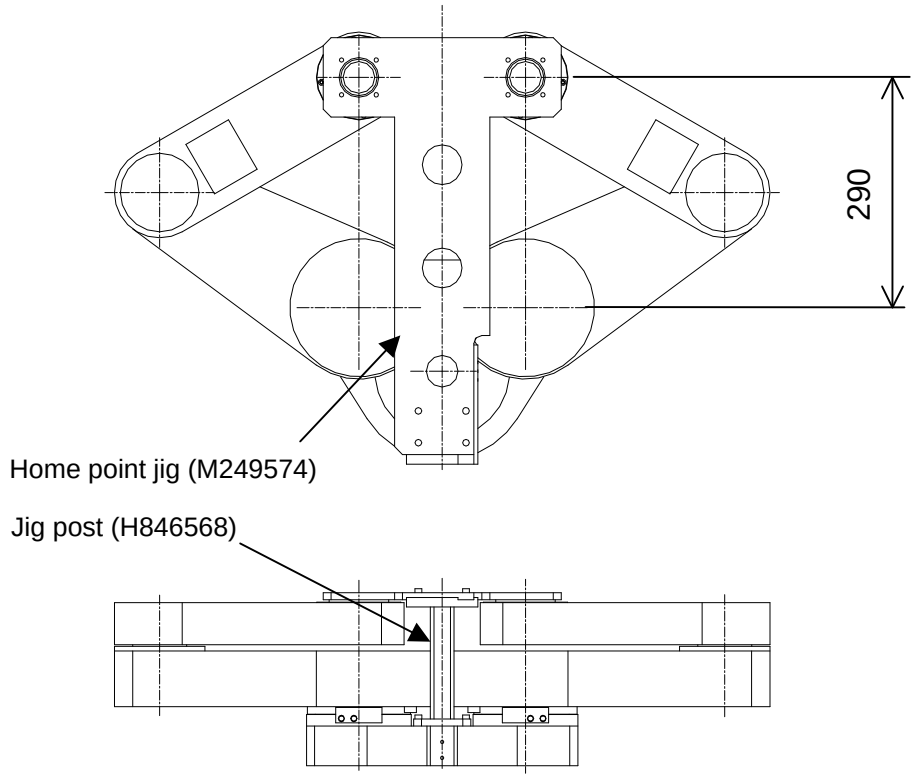
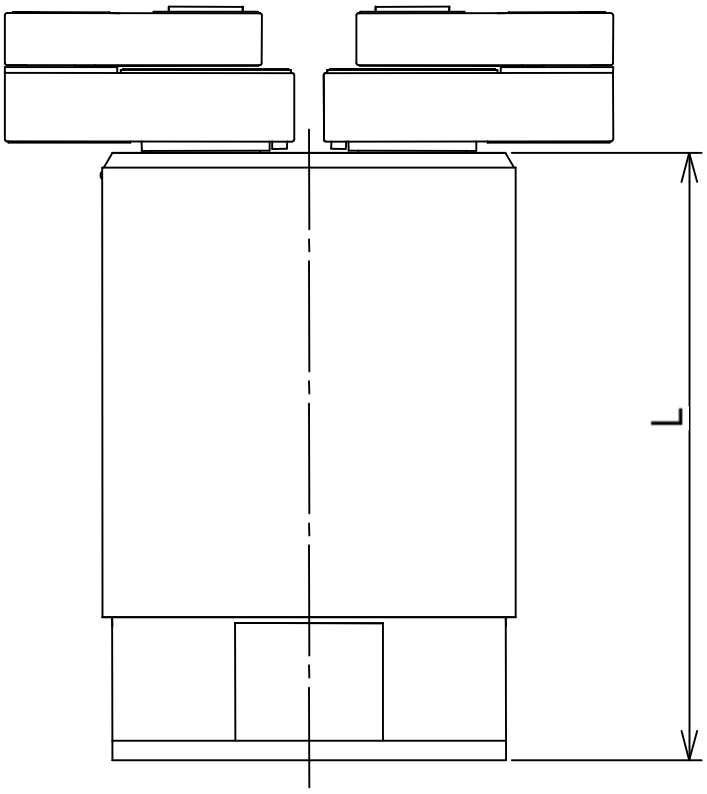


Fig. 4.1 Home point setting of arm

(2) Home point setting of axis 3 (Z-axis)

Perform the ZEROP at the following position (i.e., dimension L±0.5*), using a scale, etc.

* Dimension from the robot bottom surface to the base plate top surface: L mm



L=620(mm)

Fig. 4.2 Home point setting of axis 3 (Z-axis)

(3) Home point setting of axis 4 (TH-axis)

Set the home point by symmetrizing the distance between the mechanical stoppers. (Absolute value: Approx. 341°)

First of all, perform the ZEROP at an approximate position.

Next, move the axis in the minus (–) direction by hand. Jot down the coordinate of a position where the axis is stopped by the mechanical stopper.

Likewise, move the axis to the mechanical stopper position in the plus (+) direction and jot down the coordinate then.

The sum of these two (2) points is divided by 2 and the result is taken as the position where the ZEROP is to be executed.

Move the axis 4 to this position, then execute the ZEROP.

(4) Home point setting of axis 5 (Traverse-axis)

First of all, perform the ZEROP at an approximate position.

Next, move the axis to the mechanical stopper position in the plus (-) direction.

Execute the ZEROP at the position of 5±0.5 (mm) from the mechanical stopper, using a scale, etc.

4.2 Procedures for Re-Setting Each Axis Home Point

Set each axis at the position described in Para. 4.1 above, then perform the home point setting operation (ZEROP) in the following manner.

a) Setting of new home position data (To be set through teach pendant.)

1) Select the TEACHING mode by means of the master mode switch.

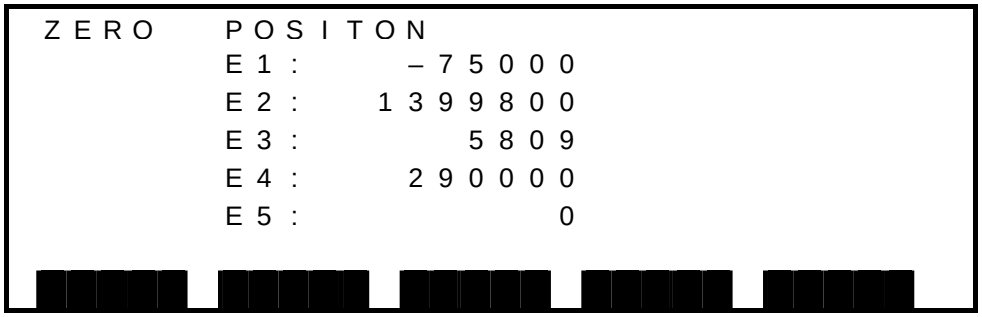
2) Press the UTILITY key on the teach pendant.

3) Press the NEXT key on the teach pendant twice.

4) Turn the servo power off.

5) Press the [F3] key on the teach pendant to select the ZEROP mode.

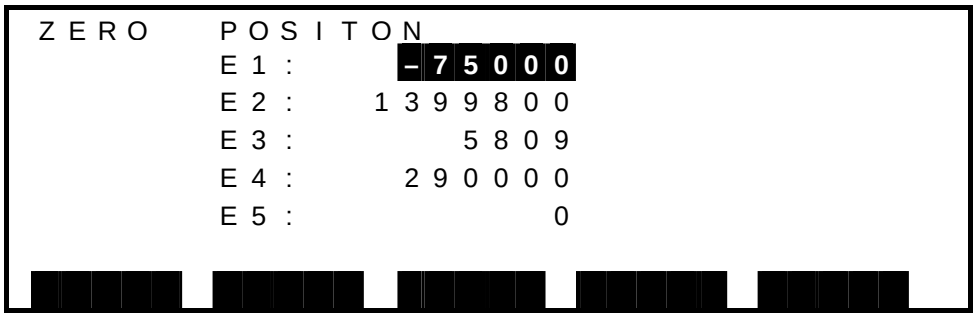
6) The following home point setting screen is called.



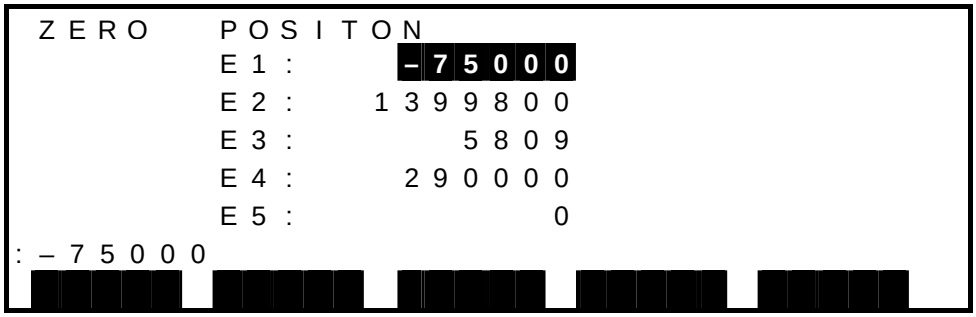
7) Press the ALT and 0 keys on the teach pendant.

8) The following home point editing screen appears.

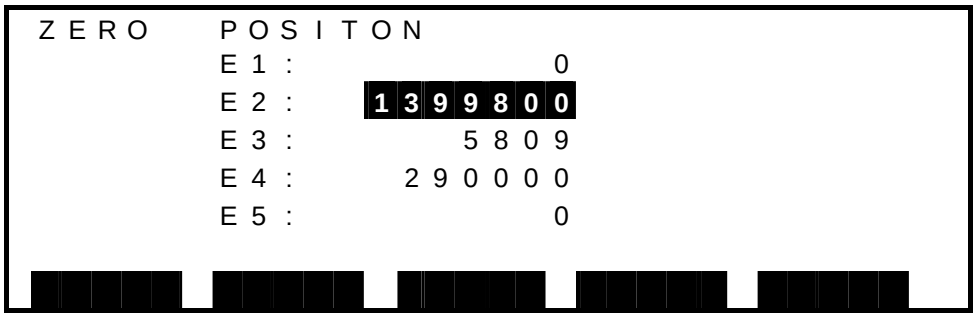
9) The numerical value of E1 (axis 1) is inversely displayed.



10) Press the EXE key on the teach pendant, and the value of "E1" is displayed at the lower left side. Set "0" in lieu of this value and press the EXE key. Then the E1 value becomes "0".
Unless the axis could be set to the home point, enter the coordinate value as recorded beforehand.



11) When the "↓" key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.



b) Confirmation of home position

1) Make sure of the robot current position according to the POS value of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position.

2) Execute the following program and make sure of the robot machine home position.

```
   PROGRAM ZERO
   SPEED=30
   MOVEA 3,100
   MOVEA 1,0
   MOVEA 2,0
   MOVEA 3,0
   MOVEA 4,0
END
```

c) Data saving

Save the home point data.
Connect the personal computer installed TSPC (option) with the controller, then back up the "MACHINE.PAR" file saved in the robot to the personal computer.
Save the "MACHINE.PAR" in the system disk.

The home point data are characteristic of each robot, which differ with the robot you use. Be sure to save the data.

Now the operation has finished.

Appendix Table 1 List of Mechanical Replacement Parts for Maintenance

Part name	Type		Maker	Q'ty	Remarks
Servo motor	VLBST-Z02030S -OU001	Y610A1D60	Toshiba Machine	2 (3)	Axis 1 (R)(, Axis 2), Axis 4 (TH)
	VLBST-Y04030S -BOU002	Y610A5630		1	Axis 3 (Z)
Reduction	SHF-25-80-2UH-SP	S727367	Toshiba Machine	1 (2)	Axis 1 (R)(, Axis 2)
gear	SHF-25-80-2SO-SP	S727368		1	Axis 4 (TH)
Ball screw	BLK2020-7.2ZZ+534L C7T	M118336	Toshiba Machine	1	Axis 3 (Z)
Linear guide	NR25XR1UUC1S+52 0LHFS-Ⅱ	M126145	Toshiba Machine	1	Axis 3 (Z)
Timing belt	935–5GT–30		Toshiba Machine	1 (2)	In the arm 1
	810–5GT–15			1 (2)	In the arm 2
	340-5GT-15			1	In the base
Grease	AFE grease		THK	–	Axis 3 (Z), traverse unit
Battery	Size AA alkali battery			3	In the Battery Box

! CAUTION

- With the exception of the grease, all parts listed above are custom-made to Toshiba Machine specifications. Contact Toshiba Machine at order entry.
- NEVER replace the parts other than the lithium battery by the user.

Appendix Table 2 List of Controller Replacement Parts for Maintenance

Part name	Type	Maker	Q'ty	Remarks
PS1 (Switching power supply)	LEB100F–0524	Cosel	1	
Main power switch	IR-11-AB-10A-1B	Nippon Thermo	1	
Lithium battery	ER3V-C	Toshiba Denchi	1	
Fuse (X8HN/X8HI printed board)	51NM030H	PICO	1	
Printed circuit board	X8HC (main control)	Toshiba Machine	1	
	X8HJ (connector)		1	
	X8HN, X8HI (I/O printed board)		1	
	X8HS (servo)		1	
	X8HP (servo power supply)		1	
	X8HBA (mother board)		1	Standard: Five (5)-axis control
Operation panel unit	X8HO	Toshiba Machine	1	
Teach pendant	TP1000	Toshiba Machine	1	With 5m-long cable
System disk	TS2000SYS	Toshiba Machine	1	

Appendix Table 3 List of Replacement Cables for Maintenance

Without Traverse axis

No.	Type of cable	Name	Q'ty	Remarks
1	KL3256G04	Cable assembly	1	I/O cable in the arm (S/W-arm common) 3 input type
2	KL3256G04	Cable assembly	2	I/O cable in the arm (S-arm) 6 input type
3	KL3256G03	Cable assembly	1	I/O cable in the arm (W-arm) 3 input type
4	KL3258G01	Cable assembly	1	I/O cable in the base
5	KL3257G01	Cable assembly	1	Power drive cable in the base
6	H07438G01	Cable assembly	1	Encoder cable in the base

(Q'ty) is Option.

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